# Finite-Sample Guarantees for Regularized RKHS Pearson Classification

## 1 Setting and classification score

Let $\mathcal{H}$ be a real separable reproducing kernel Hilbert space with measurable feature map $\phi(x) = k(x, \cdot)$.

**Assumption 1** (Kernel and sampling)**.** *The positive-semidefinite kernel satisfies*

$$k(x, y) \geq 0, \qquad k(x, x) \leq K$$

*for all $x, y$, where $K > 0$.*

*For a fixed reference regime $P_i$ and an incoming distribution $Q$, let*

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} P_i, \qquad Z_1, \ldots, Z_n \overset{\text{iid}}{\sim} Q,$$

*with the two samples independent. The kernel and the regularization parameter $\lambda > 0$ are fixed before these samples are drawn.*

For any distribution $P$, define its mean embedding and uncentered second-moment operator by

$$\mu_P = \mathbb{E}_P[\phi(X)], \qquad T_P = \mathbb{E}_P[\phi(X) \otimes \phi(X)],$$

where

$$(\phi(x) \otimes \phi(x))f = \langle \phi(x), f \rangle_{\mathcal{H}} \, \phi(x).$$

In particular,

$$\langle f, T_P f \rangle_{\mathcal{H}} = \mathbb{E}_P[f(X)^2].$$

For the reference sample, write

$$\widehat{\mu}_i = \frac{1}{n} \sum_{j=1}^{n} \phi(X_j), \qquad \widehat{T}_i = \frac{1}{n} \sum_{j=1}^{n} \phi(X_j) \otimes \phi(X_j).$$

For the incoming sample, write

$$\widehat{\mu}_Q = \frac{1}{n} \sum_{j=1}^{n} \phi(Z_j).$$

Set

$$\Delta_i = \mu_Q - \mu_{P_i}, \qquad \widehat{\Delta}_i = \widehat{\mu}_Q - \widehat{\mu}_i,$$

and

$$A_i = T_{P_i} + \lambda I, \qquad \widehat{A}_i = \widehat{T}_i + \lambda I.$$

The population and empirical scores are

$$D_{\lambda,i} := \frac{1}{2} \left\langle \Delta_i, A_i^{-1} \Delta_i \right\rangle_{\mathcal{H}}, \tag{1}$$

$$\widehat{D}_{\lambda,i} := \frac{1}{2} \left\langle \widehat{\Delta}_i, \widehat{A}_i^{-1} \widehat{\Delta}_i \right\rangle_{\mathcal{H}}. \tag{2}$$

These quantities are well-defined because $A_i, \widehat{A}_i \succeq \lambda I$.

The score is a regularized RKHS Pearson-type discrepancy. It need not equal the classical unregularized Pearson divergence.

For a pairwise comparison, the incoming window is accepted as compatible with regime $i$ when

$$\widehat{D}_{\lambda,i} \le \eta,$$

and rejected otherwise.

## 2 Null guarantee and threshold

Define, for $\rho_0 \in (0,1)$,

$$U_n(\rho_0) := \frac{K}{n\lambda}\left(1 + \sqrt{\log \frac{1}{\rho_0}}\right)^2. \tag{3}$$

**Theorem 1** (Acceptance under the null)**.** *Under Assumption 1, if $Q = P_i$, then*

$$\mathbb{P}\left\{\widehat{D}_{\lambda,i} \le U_n(\rho_0)\right\} \ge 1 - \rho_0.$$

*Consequently, any threshold $\eta \ge U_n(\rho_0)$ has pairwise false-rejection probability at most $\rho_0$.*

*Proof.* Under the null, $\Delta_i = 0$. Independence and centering give

$$\mathbb{E}\left\|\widehat{\Delta}_i\right\|_{\mathcal{H}}^2 = \frac{2}{n}\left(\mathbb{E}_{P_i}[k(X,X)] - \|\mu_{P_i}\|_{\mathcal{H}}^2\right) \le \frac{2K}{n}.$$

Hence

$$\mathbb{E}\left\|\widehat{\Delta}_i\right\|_{\mathcal{H}} \le \sqrt{\frac{2K}{n}}.$$

Nonnegativity of the kernel implies

$$\|\phi(x) - \phi(y)\|_{\mathcal{H}}^2 = k(x,x) + k(y,y) - 2k(x,y) \le 2K.$$

Replacing any one of the $2n$ observations therefore changes $\left\|\widehat{\Delta}_i\right\|_{\mathcal{H}}$ by at most $\sqrt{2K}/n$. The sum of squared bounded-difference constants is $4K/n$.

One-sided McDiarmid concentration yields

$$\mathbb{P}\left\{\left\|\widehat{\Delta}_i\right\|_{\mathcal{H}} > \sqrt{\frac{2K}{n}}\left(1 + \sqrt{\log \frac{1}{\rho_0}}\right)\right\} \le \rho_0.$$

Finally, $\widehat{A}_i^{-1} \preceq \lambda^{-1} I$ gives

$$\widehat{D}_{\lambda,i} \le \frac{1}{2\lambda}\left\|\widehat{\Delta}_i\right\|_{\mathcal{H}}^2,$$

which proves the result. □

## 3 Directional guarantee under the alternative

The alternative proof controls the empirical score along one fixed population direction. It requires concentration of two scalar averages rather than an operator-norm bound for $\widehat{T}_i - T_{P_i}$.

### 3.1 Variational representation and population direction

For any bounded self-adjoint operator $B \succeq \lambda I$ and $v \in \mathcal{H}$, completing the square gives

$$\frac{1}{2}\left\langle v, B^{-1}v\right\rangle_{\mathcal{H}} = \sup_{f\in\mathcal{H}}\left\{\langle f, v\rangle_{\mathcal{H}} - \frac{1}{2}\langle f, Bf\rangle_{\mathcal{H}}\right\}. \tag{4}$$

Indeed,

$$\langle f, v\rangle_{\mathcal{H}} - \frac{1}{2}\langle f, Bf\rangle_{\mathcal{H}} = \frac{1}{2}\left\langle v, B^{-1}v\right\rangle_{\mathcal{H}} - \frac{1}{2}\left\langle f - B^{-1}v, B(f - B^{-1}v)\right\rangle_{\mathcal{H}}.$$

Suppose $D_{\lambda,i} > 0$ and define

$$u_i := \frac{A_i^{-1}\Delta_i}{\sqrt{2D_{\lambda,i}}}. \tag{5}$$

This direction depends only on the population distributions, the kernel, and $\lambda$. It does not depend on the observed samples.

Its normalization gives

$$\langle u_i, \Delta_i\rangle_{\mathcal{H}} = \sqrt{2D_{\lambda,i}}, \tag{6}$$

$$\langle u_i, A_i u_i\rangle_{\mathcal{H}} = 1. \tag{7}$$

Since $A_i \succeq \lambda I$, it follows that

$$\|u_i\|_{\mathcal{H}}^2 \le \frac{1}{\lambda}. \tag{8}$$

Define

$$\widehat{d}_i := \left\langle u_i, \widehat{\Delta}_i\right\rangle_{\mathcal{H}} = \frac{1}{n}\sum_{j=1}^{n} u_i(Z_j) - \frac{1}{n}\sum_{j=1}^{n} u_i(X_j), \tag{9}$$

$$\widehat{q}_i := \left\langle u_i, \widehat{A}_i u_i\right\rangle_{\mathcal{H}} = \frac{1}{n}\sum_{j=1}^{n} u_i(X_j)^2 + \lambda\|u_i\|_{\mathcal{H}}^2. \tag{10}$$

Their expectations satisfy

$$\mathbb{E}\widehat{d}_i = \sqrt{2D_{\lambda,i}}, \qquad \mathbb{E}\widehat{q}_i = 1.$$

Applying (4) to the empirical score and restricting the supremum to $f = tu_i$, $t \ge 0$, gives

$$\begin{aligned}\widehat{D}_{\lambda,i} &\ge \sup_{t\ge 0}\left\{t\widehat{d}_i - \frac{t^2}{2}\widehat{q}_i\right\} \\ &= \frac{\left[\widehat{d}_i\right]_+^2}{2\widehat{q}_i},\end{aligned} \tag{11}$$

where $[x]_+ = \max\{x, 0\}$. The denominator is strictly positive because $u_i \neq 0$ and $\widehat{A}_i \succeq \lambda I$.

Thus, a lower bound on the empirical score follows from a lower bound on $\widehat{d}_i$ and an upper bound on $\widehat{q}_i$.

### 3.2 Concentration of the two scalar quantities

Fix $\rho_1 \in (0, 1)$ and set

$$\ell_1 := \log\frac{2}{\rho_1}, \qquad s_n(\rho_1) := \sqrt{\frac{K\ell_1}{n\lambda}}, \qquad r_n(\rho_1) := \frac{K}{\lambda}\sqrt{\frac{\ell_1}{2n}}. \tag{12}$$

**Directional mean difference.** The reproducing property and the nonnegative kernel give

$$|u_i(x) - u_i(y)| \le \|u_i\|_{\mathcal{H}} \|\phi(x) - \phi(y)\|_{\mathcal{H}} \le \sqrt{2K}\, \|u_i\|_{\mathcal{H}}\,.$$

Therefore, each of the $2n$ independent summands in (9) has range width at most $\sqrt{2K}\, \|u_i\|_{\mathcal{H}}\, /n$.

One-sided Hoeffding concentration yields, for $t > 0$,

$$\mathbb{P}\left\{\widehat{d}_i - \mathbb{E}\widehat{d}_i \le -t\right\} \le \exp\left(-\frac{nt^2}{2K\, \|u_i\|_{\mathcal{H}}^2}\right).$$

Using (8), we obtain

$$\widehat{d}_i \ge \sqrt{2D_{\lambda,i}} - \sqrt{2}\, s_n(\rho_1) \tag{13}$$

with probability at least $1 - \rho_1/2$.

**Directional quadratic term.** For every $x$,

$$0 \le u_i(x)^2 \le k(x,x)\, \|u_i\|_{\mathcal{H}}^2 \le K\, \|u_i\|_{\mathcal{H}}^2\,.$$

Only the empirical average in (10) is random. Hoeffding concentration therefore gives

$$\mathbb{P}\left\{\widehat{q}_i - 1 \ge t\right\} \le \exp\left(-\frac{2nt^2}{K^2\, \|u_i\|_{\mathcal{H}}^4}\right).$$

Using (8) again gives

$$\widehat{q}_i \le 1 + r_n(\rho_1) \tag{14}$$

with probability at least $1 - \rho_1/2$.

The two events need not be independent: they share the reference sample. A union bound ensures that (13) and (14) hold simultaneously with probability at least $1 - \rho_1$.

### 3.3 Alternative guarantee

**Assumption 2** (Population separation)**.** *For the fixed pair $(Q, P_i)$,*

$$D_{\lambda,i} \ge \delta > 0.$$

**Theorem 2** (Directional lower bound)**.** *Under Assumptions 1 and 2, define*

$$L_n(\delta, \rho_1) := \frac{\left[\sqrt{\delta} - s_n(\rho_1)\right]_+^2}{1 + r_n(\rho_1)}. \tag{15}$$

*Then*

$$\mathbb{P}\left\{\widehat{D}_{\lambda,i} \ge L_n(\delta, \rho_1)\right\} \ge 1 - \rho_1.$$

*In particular, if $\eta < L_n(\delta, \rho_1)$, then*

$$\mathbb{P}\left\{\widehat{D}_{\lambda,i} > \eta\right\} \ge 1 - \rho_1.$$

*Proof.* On the joint event established above, (11) gives

$$\widehat{D}_{\lambda,i} \ge \frac{\left[\sqrt{2D_{\lambda,i}} - \sqrt{2}\, s_n(\rho_1)\right]_+^2}{2(1 + r_n(\rho_1))} = \frac{\left[\sqrt{D_{\lambda,i}} - s_n(\rho_1)\right]_+^2}{1 + r_n(\rho_1)}.$$

The right-hand side is nondecreasing in $D_{\lambda,i}$. Substituting $D_{\lambda,i} \ge \delta$ proves the claim. □

**Remark 1** (Role of the direction)**.** *The direction $u_i$ is used only in the proof. The classifier still evaluates the full empirical score* (2)*; it does not need to know or estimate $u_i$.*

*The separation assumption concerns the regularized Pearson-type score. No additional lower bound on $\|\mu_Q - \mu_{P_i}\|_{\mathcal{H}}$ is required.*

# 4 Threshold calibration and nonempty interval

## 4.1 Separate null and alternative error levels

The two guarantees are compatible whenever a threshold can be chosen in

$$U_n(\rho_0) \le \eta < L_n(\delta, \rho_1). \tag{16}$$

Because $U_n(\rho_0) > 0$, this interval is nonempty if and only if

$$\delta > \left[ s_n(\rho_1) + \sqrt{(1 + r_n(\rho_1)) U_n(\rho_0)} \right]^2. \tag{17}$$

Indeed, $U_n(\rho_0) < L_n(\delta, \rho_1)$ first requires $\sqrt{\delta} > s_n(\rho_1)$, after which taking square roots and rearranging gives (17).

The choice

$$\eta = U_n(\rho_0)$$

then ensures:

- under $Q = P_i$, acceptance with probability at least $1 - \rho_0$;
- under $D_{\lambda,i} \ge \delta$, rejection with probability at least $1 - \rho_1$.

These are guarantees under two different hypotheses, not a single joint probability statement.

Condition (17) is equivalent to nonemptiness of the certified interval (16). It is only a sufficient condition for successful classification itself.

## 4.2 Explicit formulas in the initial parameters

Taking $\rho_0 = \rho_1 = \rho$, choose

$$\boxed{\eta = \frac{K}{n\lambda} \left( 1 + \sqrt{\log \frac{1}{\rho}} \right)^2.} \tag{18}$$

The required population separation is

$$\boxed{\delta > \frac{K}{n\lambda} \left[ \sqrt{\log \frac{2}{\rho}} + \left( 1 + \sqrt{\log \frac{1}{\rho}} \right) \sqrt{1 + \frac{K}{\lambda} \sqrt{\frac{\log(2/\rho)}{2n}}} \right]^2.} \tag{19}$$

For an independently chosen threshold $\eta \ge U_n(\rho)$, the corresponding sufficient condition is

$$\boxed{\delta > \left[ \sqrt{\frac{K \log(2/\rho)}{n\lambda}} + \sqrt{\eta \left( 1 + \frac{K}{\lambda} \sqrt{\frac{\log(2/\rho)}{2n}} \right)} \right]^2.} \tag{20}$$

For kernels satisfying $k(x, x) = 1$, take $K = 1$. For fixed $K$, $\lambda$, and $\rho$, both the threshold in (18) and the sufficient separation in (19) are of order $n^{-1}$.

### 4.3 Numerical illustration

For

$$n = 100, \qquad K = 1, \qquad \lambda = 0.1, \qquad \rho = 0.1,$$

the quantities above are approximately

$$U_n(\rho) = 0.63374, \qquad s_n(\rho) = 0.54733, \qquad r_n(\rho) = 1.22387.$$

Thus the threshold $\eta = 0.63374$ is certified whenever

$$\delta > 3.00849.$$

For the same $K$, $\lambda$, and $\rho$, increasing the window length to $n = 300$ gives

$$\eta = 0.21125, \qquad \delta > 0.83985.$$

These values specify sufficient separations; they do not establish that a particular collection of regimes satisfies them.

## 5 Scope and implementation considerations

The results above concern a fixed pair of distributions and fixed sampling windows.

Reference-specific kernels are permitted: for each comparison, the same reference kernel must be used to construct both mean embeddings and the reference second-moment operator. Each kernel must satisfy the stated assumptions. A kernel learned from an independent preliminary sample can be handled by conditioning on that sample, provided the kernel bound holds uniformly.

A bandwidth estimated from the same reference observations used in the score requires an additional argument; freezing it after estimation does not by itself establish the fixed-kernel sampling assumptions.

For a memory containing several regimes, correct assignment requires control of the relevant comparisons simultaneously. Likewise, repeated online decisions and windows selected by a change-point detector require appropriate treatment of multiple comparisons, selection, dependence, and possible regime contamination. The pairwise results alone do not establish those guarantees.

Finally, the implementation must evaluate (2) with the same normalization and regularization convention. Thresholds calibrated for a different empirical quadratic score are not directly covered by these results.